

Automating Tasks

Objectives

After completing this lesson, you should be able to:

- Describe the available automated task utilities
- Configure `cron` jobs
- Describe `cron` directories and files
- Use the `crontab` utility
- Configure `anacron` jobs
- Use the `at` and `batch` utilities



Automating System Tasks

- Oracle Linux can run tasks automatically, and comes with automated task utilities such as `cron`, `anacron`, `at`, and `batch`.
- `cron` jobs can run as often as every minute.
 - A scheduled `cron` job is skipped if the system is down.
- `anacron` can run a job only once a day.
 - Scheduled jobs are remembered and run the next time the system is up.
- `crond` searches multiple files and directories for scheduled jobs:
 - `/var/spool/cron/`
 - `/etc/anacrontab`
 - `/etc/cron.d`



Configuring `cron` Jobs

- `cron` jobs are defined in `/etc/crontab`.
- Jobs are defined by:
 - minute: From 0 to 59
 - hour: From 0 to 23
 - day: From 1 to 31
 - month: From 1 to 12, or the name of the month
 - day-of-week: From 0 to 7, or the abbreviated name of day
 - command: Program or script to execute
- Special characters can be used:
 - Asterisk, hyphen, comma, and forward slash



Other `cron` Directories and Files

- `/etc/cron.d`
 - Contains files with the same syntax as `/etc/crontab`
 - `root` privileges only
- Other `cron` directories in `/etc`:
 - `cron.hourly`
 - `cron.daily`
 - `cron.weekly`
 - `cron.monthly`
- Scripts in these directories run hourly, daily, weekly, or monthly, depending on the name of the directory.
- The `/etc/cron.allow` and `/etc/cron.deny` files restrict user access to `cron`.

The `crontab` Utility

- The `crontab` utility allows users other than `root` to configure `cron` tasks.
- User-defined `crontabs` are stored in:
 - `/var/spool/cron/<user>`
- To create or edit a `crontab`:
 - Enter the following command:

```
$ crontab -e
```

- Use the same format as `/etc/crontab` without specifying a user.

- To list the contents of a user-defined `crontab`, enter:

```
$ crontab -l
```

- The `root` user can use the `-u <username>` option to create, edit, remove, or list the `crontab` for a specific user. For example:

```
# crontab -u oracle -l
```

Configuring `anacron` Jobs

- `anacron` jobs are defined in `/etc/anacrontab`.
- Jobs are defined by:
 - Period in days: The frequency of execution in days
 - Delay in minutes: The minutes to wait before executing the job
 - Job-identifier: A unique name used in log files
 - Command: A shell command or script to execute



Checking cron Job Status

- cron messages logs are specified in `/etc/rsyslog.conf`.
- Sample `/etc/rsyslog.conf` entry that logs all cron messages to `/var/log/cron`:

```
# Log cron stuff
cron.*                /var/log/cron
```

- Information about cron and anacron jobs are stored here.
- Sample log entries:

```
Aug  7 09:01:01 host03 CROND[5607]: (root) CMD (run-parts
/etc/cron.hourly)
Aug  7 09:01:01 host03 run-parts (/etc/cron.hourly) [5607]:
starting 0anacron
Aug  7 09:01:01 host03 run-parts (/etc/cron.hourly) [5618]:
finished 0anacron
```


at and batch

- `at` and `batch` are utilities for scheduling one-time tasks.
- The `at` command executes a task at a specific time.
- The `batch` command executes a task when the system load average drops below 0.8.
- The `atd` service must be running to use `at` or `batch`.
- `at` command syntax:
 - `at time`
 - The `time` argument is the time to execute the command.
 - The `time` argument accepts multiple formats.
- `batch` command syntax:
 - `batch` (the `at>` prompt is displayed)



Quiz



Both `at` and `batch` are used to schedule the execution of recurring tasks.

- a. True
- b. False



Summary

In this lesson, you should have learned how to:

- Describe the available automated task utilities
- Configure `cron` jobs
- Describe `cron` directories and files
- Use the `crontab` utility
- Configure `anacron` jobs
- Use the `at` and `batch` utilities



Practice 8: Overview

This practice covers automating tasks by using the `crontab` and `at` commands.